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Wearable light spectral sensor optimized for measuring daily α-opic light exposure

ANAS MOHAMED,[1] VINEETHA KALAVALLY,[1,*] SEAN W. CAIN,[2] ANDREW J. K. PHILLIPS,[2] ELISE M. MCGLASHAN,[2] AND CHEE PIN TAN[3]

[1]*Electrical Engineering and Computer Systems Engineering Department and Intelligent Lighting Laboratory, Monash University, Bandar Sunway, 47500 Selangor, Malaysia*
[2]*School of Psychological Sciences and Turner Institute for Brain and Mental Health, Monash University, VIC, Australia*
[3]*Robotics and Mechatronics Engineering Department and Advanced Engineering Platform, Monash University, Bandar Sunway, 47500 Selangor, Malaysia*
[*]*vineetha@monash.edu*

Abstract: Light has many non-visual effects on human physiology, including alterations in sleep, mood, and alertness. These effects are mainly mediated by photoreceptors containing the photopigment melanopsin, which has a peak sensitivity to short wavelength ('blue') light. Commercially available light sensors are commonly wrist-worn and report photopic illuminance and are calibrated to perceive visual brightness and hence cannot be used to investigate the non-visual impacts of light. In this paper, we report the development of a wearable spectrophotometer designed to be worn as a pendant or affixed to clothing to capture spectral power density data close to eye level in the visible wavelength range 380-780 nm. From this, the relative impact of a given light stimulus can be determined for each photoreceptive input in the human eye by calculating effective illuminances. This device showed high accuracy for all effective illuminances while measuring a range of commonly encountered light sources by calibrating for directional response, dark noise, sensor saturation, non-linearity, stray-light and spectral response. Features of the device include IoT-integration, onboard data storage and processing, Bluetooth Low Energy (BLE) enabled data transfer, and cloud storage in one cohesive unit.

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1. Introduction

Light enables image-forming vision by stimulating the S-, M-, L-cones, and the rods, which are the classical photoreceptors for vision in the human eye. Light also drives 'non-visual' effects which are mediated by intrinsically photosensitive retinal ganglion cells (ipRGCs) that contain a fifth photopigment, melanopsin, and also receive inputs from the classical photoreceptors. All five photoreceptors have distinct peak spectral sensitivities (Supplement 1, Fig. S1) that need to be measured to accurately determine the visual and non-visual effects of light [1].

The non-visual effects of light include pupillary light responses, suppressing the sleep-promoting hormone melatonin, and synchronizing the central circadian clock [2–5]. In line with the peak sensitivity of melanopsin (~479nm) [6], action spectra for melatonin suppression demonstrate a short-wavelength peak sensitivity (λ_{max} ~460-480 nm; [7–10]) that does not match photopic (λ_{max} 555 nm) or scotopic (λ_{max} 510 nm) vision [8,9]. Recently, the International Commission on Illumination (CIE) defined spectral sensitivity functions to describe the ability of optical radiation to stimulate each of the five photoreceptor types [1]. Measuring illuminance relative to its impact on different photoreceptors, in conjunction with the behavioral, physiological, and activity profile of a person will give a richer understanding of the visual and non-visual effects of light. This is essential for recommending lighting design which optimizes visual and non-visual functions [11,12].

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Accurate determination of relative illuminance based on the CIE spectral sensitivity functions requires the collection of high-resolution spectral power density (SPD) data. However, wearable light measurement devices typically only collect photopic illuminance, or red, green, blue (RGB) light data. Conventional devices which measure high-resolution SPD of light are primarily designed to be used in a laboratory setting, making them impractical and expensive for field research involving measurements of daily light exposure of people in ecologically valid settings. Where attempts have been made to evaluate the effects of light on the circadian system with wearables [13–15], typically a simple photodiode with filters has been used to approximate at most the photopic and melanopic illuminances.

This work aims to establish viable ways and motivate standards (which currently do not exist) for the metrology of optical radiation targeting human photoreception using wearable sensors. The device designed to be used in both laboratory and field settings measures and records SPD data and has the onboard capacity to calculate equivalent daylight illuminances (EDIs) based on CIE S026:2018. The characterization and calibration of this device based on CIE 233: 2019 and the procedures used to develop it are described in this paper, along with the experimental outcomes.

2. Design, characterization and calibration the device

2.1. Conceptual design

A radiometric measurement of light such as irradiance is expressed in terms of its SPD, defined as the radiant flux received by a surface per unit area per nm wavelength. The corresponding photometric measurement, photopic illuminance, is calculated by weighting the radiant flux density by the luminous efficiency function, which describes the average spectral sensitivity of the human visual perception of brightness. The input to the human non-visual system on the other hand can be represented by an alternative measurement system initially proposed by Lucas et. al. [16]and more recently adapted for a CIE standard [1], which enables calculations of α-opic equivalent daylight illuminances (α-opic EDIs) corresponding to the five photoreceptors found in the human eye. Thus, an instrument designed for measuring and recording the efficiency of optical radiation on all the five photoreceptors should incorporate into its design a set of action spectra (shown in Supplement 1, Fig. S1), in addition to the measured SPD.

The most common method to obtain spectrally-weighted responses from sensors is to use a system of photodiodes and filters to limit the absorption bands in the visual range such that they closely approximate the spectral response function of interest [13,14]. This method is often inaccurate due to a mismatch between the final absorption range of the photodiode-filter system and the action spectrum of the photoreceptor. Previous attempts to achieve this only do so with at most two of the efficiency functions of the human eye [14]. This most commonly includes the photopic luminous efficiency function and the melanopic efficiency function. The human photoreceptors have unique functions and likely contribute in different ways to the non-visual effects of light. However, the evaluation of their contribution requires an appropriate sensor that can determine the effect of a light environment on all photoreceptors. Therefore, in our design, we propose a novel approach where, like a spectrophotometer, we measure the SPD of the entire visual range, after which the action spectra for each of the photoreceptors $s_\alpha(\lambda)$ and $V(\lambda)$ are used to compute the five α-opic EDIs and the photopic illuminance, respectively. The α-opic EDI is defined as the illuminance produced by radiation conforming to a standard daylight (D65) that provides equal α-opic irradiance as the test source expressed by $E_{v,\alpha}^{D65} = E_v.\gamma_{\alpha,v}^{D65}$, where $\gamma_{\alpha,v}^{D65}$ (α-opic daylight efficacy ratio) is the ratio of the α-opic efficacy of luminous radiation for a source to that for D65 daylight [1].

Since the timing and duration of radiation exposure also affects non-visual responses [17], it is equally important to record daily light exposure patterns as a time series as it is to record the

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SPD with the highest plausible accuracy. To achieve an accurate estimate of the α-opic EDI, the device must have the following metrological performances: (i) visible light (380-780 nm) spectral sensing (ii) calibrated SPD output with a spectral resolution at least 20 nm to match commercially available mini spectral sensors (iii) a wavelength pitch of 5nm to correspond to that of the photoreceptor action spectra specified in CIE S026:2018 (iv) sufficient processing power to compute the α-opic quantities from the SPD (v) sufficient memory to store measured data for at least 5 days (typical work week) at an interval of 1 measurement per minute which is normal for such measurements with timestamps for the data collected (vi) mobility and Bluetooth connectivity and (vii) a wearable form factor.

We selected a mini-spectrophotometer (C12666MA) with 256 CMOS sensing elements coupled with a diffraction grating system from Hamamatsu Photonics Co., Ltd [18]. It was fitted with the electronic and communication peripherals required to calibrate, measure, store, and transmit spectral data and the corresponding α-opic EDIs as illustrated in Fig. 1 (a) and (b). A diffusing layer disperses incident light onto the aperture of the C12666MA which is controlled by the nRF microprocessor. The raw output of the C12666MA is then passed on to the microprocessor for processing. A multilevel calibration of the raw input is performed on the device (to be worn as shown in Fig. 2(c)) resulting in an absolute measure of the irradiance of the incident light, which is used to compute the α-opic EDIs. The calculated values and the calibrated output are stored on an on-board flash storage which can then be uploaded to a permanent cloud-based storage via an Android application.

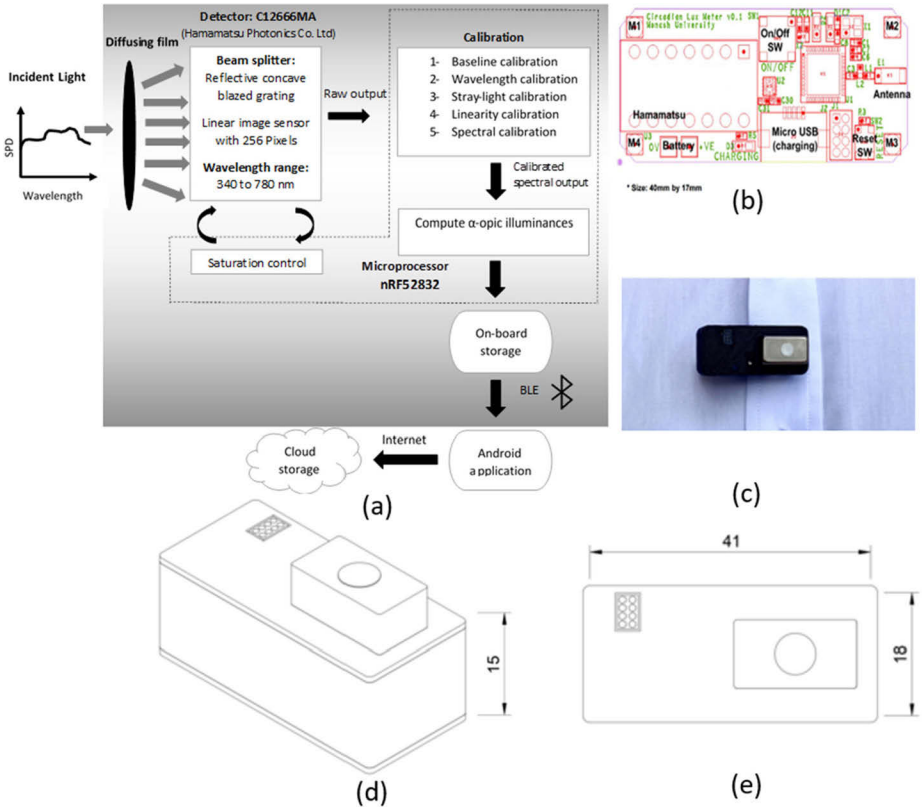


Fig. 1. (a) Schematic of the wearable spectrophotometric sensor system; (b) Schematic of the circuit board with key components; (c) Full device package; (d) Isometric view of the device; (e) Dimensions of the device.

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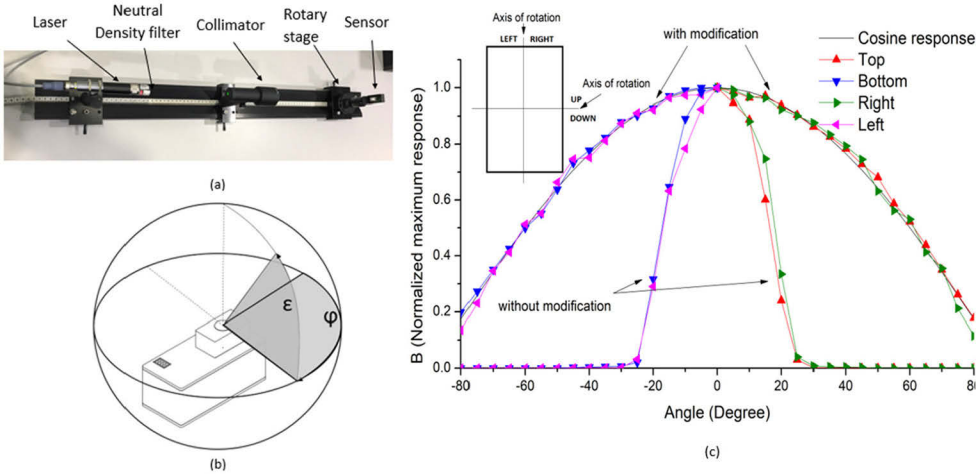


Fig. 2. (a) Setup used to investigate directional response; (b) Demonstration of the angles ε, φ; (c) Directional response of the device compared with the ideal cosine response before and after optical modification from $\varepsilon = 0$ to 80 degrees at $\varphi = 0$, 90, 180 and 270.

2.2. *Characterization and calibration*

Calibration ensures that the signal recorded by a spectral sensor is proportional to the amount of light entering the input optics in each wavelength band. Calibration compensates for sensor characteristics such as dark signal, stray-light, nonlinearity, detector saturation, and spectral response (including wavelength and irradiance calibration). Also, the device must be optically corrected to achieve a cosine directional response to mimic the spatial response of the human eye. This section describes the calibration methods that lead to accurate spectral irradiance measurements used to calculate the effective illuminances.

2.2.1. Directional response

Since the wearable sensor is intended for studying the impact of light on photoreceptors in the human eye, its spatial response must also be similar to that of the eye, which is Lambertian [19]. Thus, the directional response of the device was characterized and optically modified to be closer to Lambertian using multi-layer diffusion films to achieve a wide diffusion angle range. To investigate the directional response, the sensor was exposed to radiation from a green 532-nm laser (Z1M18B-F-532-pz, Z-LASER, Germany), the beam intensity of which was attenuated by placing a neutral density filter (EDMUND OPTICS C-Mounted Absorptive ND Filter) between the laser beam and the sensor as shown in Fig. 2(a). The laser beam was expanded and collimated to form a diameter of 10 mm. The sensor was then placed on a rotary stage (EDMUND OPTICS) and measurements made at 5-degree intervals along the two halves of horizontal and vertical semi-axes from 0 to 80 degrees.

We apply the CIE directional response mismatch [20] given by:

$$f_2(\varepsilon, \varphi, \lambda) = \frac{Y(\varepsilon, \varphi, \lambda)}{Y(0, \varphi, \lambda)\cos(\varepsilon)} - 1, \tag{1}$$

where $Y(\varepsilon, \varphi, \lambda)$ is the response of the device as a function of the two angles, the angle from the normal to the surface of the detector ε and the azimuth angle φ (shown in Fig. 2(b)). The

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directional response error aggregated over $\varepsilon_{\max}$ is given by:

$$f_2(\varphi, \lambda) = \int_0^{\varepsilon_{\max}} |f_2(\varepsilon, \varphi, \lambda)| \ \sin 2\varepsilon \ d\varepsilon, \tag{2}$$

where $\varepsilon_{\max} = 80$ degrees. The weighting factor proportional to $\sin 2\varepsilon \ d\varepsilon$ was used to define where the mismatch is considered most relevant. The directional response index is given by:

$$f_2(\lambda) = \frac{1}{4} \sum_{j=0}^{3} f_2(\varphi, \lambda) \Big|_{\varphi_j - j\frac{\pi}{2}} . \tag{3}$$

The directional response of the device is compared with the cosine response as shown in Fig. 2(c) before and after optical modifications.

The C12666MA with no optical modification has a poor directional response, sharply falling from 10 to 25 degrees angle of incidence. With the diffusion film on the optical input aperture of the sensor, its directional response closely follows a cosine (Lambertian cosine law) response along both halves of the two semi-axes. The f_2 directional response errors and index are summarized in Table 1 for both configurations.

Table 1. Sensor directional response errors and index.

f2	Without optical modification	With optical modification
Top	51.37	1.11
Bottom	51.31	0.49
Left	50.89	0.57
Right	50.51	0.42
$f_{2\ \text{index}}$	51.02	0.65

A directional response index of 0.65 is observed, which outperforms most commercially available wearable devices, and is comparable to commercial illuminance meters [15].

2.2.2. Baseline signal

Most array spectrophotometers have a baseline signal that is present even without any light input, which must be characterized and subtracted from measurements. Its two components, an electronic offset and a dark current [21], are distinguished by virtue of their dependence on the integration time of the spectrophotometer. The integration time which is the duration for which the sensor collects light input for a single measurement, is set to be long for low light levels and vice versa for high light levels for optimal measurements. The electronic offset, an outcome of mapping the sensor output voltage to the Analog-to-Digital Converter's (ADC) input voltage range, is independent of the integration time, whereas the dark current is heavily influenced by it. While the electronic offset can be assumed to be constant in stable temperature, dark current increases with integration time in complex ways, due to heating effects of purging readout cycles between measurements and charge accumulation [22]. To investigate this, repeated baseline signal measurements were taken at several integration times with the sensor in complete darkness at an ambient temperature of 25 °C. We observe higher variation in dark signal between wavelengths at longer integration times. A second order polynomial fit was applied, where the y-intersection defines the electronic offset (344 counts) of the device, which is subsequently used to predict the baseline signal for a given integration time of the sensor. Although the operating range of the sensor is between 5 °C and 50 °C, the device is optimally used at close proximity of 25 °C to achieve the reported accuracy as the output of the ADC increases or decreases

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with temperature rise and fall respectively. A temperature-based correction is possible with the inclusion of an onboard temperature sensor, which does not exist in the reported version of the device. The sensor is hermetically sealed for high humidity resistance and its response time is dependent on a dynamic setting of the sensor integration time.

2.2.3. Non-linearity calibration

Two forms of nonlinearity have to be considered for irradiance calibration, one related to the amplitude response of the sensor to irradiance and the other to the integration time of the sensor. To investigate the nonlinearity of the device with respect to the amplitude response, measurements recorded from the device were directly compared to that from a laboratory standard calibrated illuminance spectrophotometer (CL-500A Konica Minolta) with irradiance varied from 0 to 40 W/m^2. All measurements were made at an ambient temperature of 25 °C, which is within the operating temperature ranges of both the CL-500A (-10 to 40 °C) and the sensing component (5 to 50 °C). Three sets of measurements were made with the integration time of the device fixed to 300 ms and 1000 ms for the first two measurements, and a dynamic integration time setting for the last measurement. For each set of measurements, the device and the CL-500A were placed next to each other in an enclosed blackout box with the distance between the light input aperture of the CL-500A and our device at 2 cm. The intensity of the light from a programmable light source, Light Replicator (LR) from Telelumen LLC, was then progressively increased.

The results of the first two sets of measurements are plotted and compared with an ideal linear response in Fig. 3(a). These measurements clearly display the heavy dependence of the dynamic range on the integration time of the sensor. The dynamic range is worse (saturates at ~4 W/m^2) at the longer integration time (1000 ms) compared to the shorter (300 ms) integration time (saturates at ~12 W/m^2), in accordance with the expected decline in the signal-to-noise ratio (SNR) at longer integration times at a given irradiance. A shorter integration time allows for high irradiance measurements to be made before the sensor saturates but results in poor SNR performance in low light. A longer integration time causes the sensor to saturate at a lower irradiance level, but the performance is much better in low light levels. Therefore, allowing the integration time to be dynamic, depending on the light intensity, would greatly improve the dynamic range of the device. We dynamically change the integration time of the device depending on the irradiance level of the measured light, such that the SNR remains high without saturating the sensor. This process is outlined in Supplement 1, Algorithm S1.

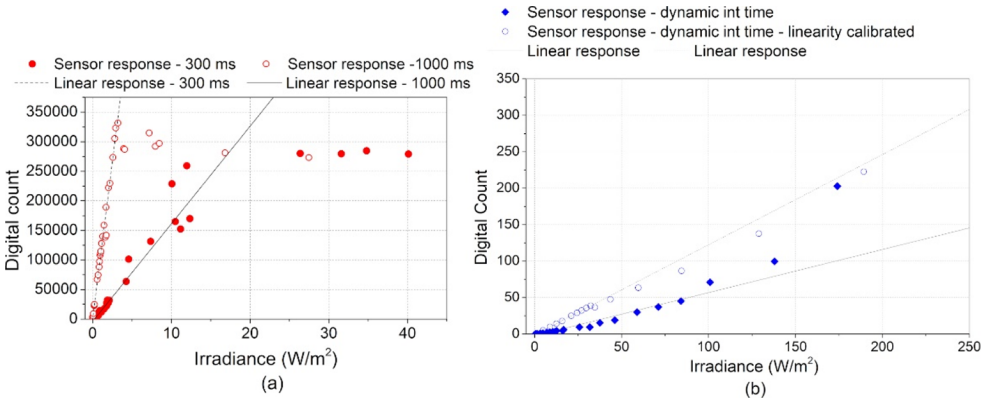


Fig. 3. Linearity characterization of the device related to the amplitude response of the sensor with (a) constant integration times of 300 ms and 1000 ms; and (b) dynamic integration time with linearity calibration.

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The third set of measurements with the device were made with this algorithm in place, where the integration time is dynamic. The response of the device, normalized with the integration time specific to each measurement (gain normalization), is plotted in Fig. 3(b). The dynamic integration time allows the linear range of the device to be greatly improved up to irradiances of $\approx$Wm^{-2}, beyond which nonlinearity was still observed. This was characterized and the nonlinearity was corrected, which allowed for an even higher linear range (190 Wm^{-2}) of the device as shown in Fig. 3(b). Beyond this, the integration time of the device could not be reduced any further and can be concluded as the saturation point of the sensor.

2.2.4. Wavelength calibration

Wavelength and irradiance calibration jointly quantify the SPD of light incident on a spectral sensor. Sixteen channels that are distributed within the 395-735 nm wavelength range, from the Telelumen Light Replicator (LR), were used to characterize the wavelength accuracy of the device using calibration data provided by the manufacturer of the spectral sensing unit (C12666MA). The application of this wavelength calibration resulted in a wavelength pitch between ~1.2 - 6nm with a manufacturer reported spectral resolution of 15 nm. The channels of the LR were turned on one at a time. The CL-500A with a spectral range of 360-780 nm, spectral resolution of 10 nm, wavelength pitch of 1 nm and a wavelength precision of $\pm$ 0.3 nm and our device were placed next to each other in an enclosed black out box. The LR is fixed to the top of box with a set distance of 1 m between the sensors and the light source. The distance between the light input aperture of the CL-500A and our device was at 2 cm. Twenty repeated measurements were then taken with both the standard spectrophotometer and the test device, concurrently at 15 s intervals. The measurement procedure is intended to minimize any external light influence and variations in incident peak wavelength between devices. The mean of the peak wavelengths recorded using the CL-500A and our device revealed a wavelength error within ±6 nm in the measurement range.

2.2.5. Stray light

Next, we describe how stray-light correction for regular spectrophotometers was adapted to calibrate the wearable device. Stray light is light that reaches the detector at wavelengths other than the wavelength intended to be measured [22]. It can be caused by in-range light scattering from components such as gratings, mirrors, etc. and out-of-range light outside the spectral range of the device. Its effects are most pronounced when a spectrophotometer is exposed to a broadband source with higher radiation in the Near Infrared Range (NIR) range relative to its radiation in the visible range [23]. To investigate the effects of stray light, the sensor was placed in a light booth (SpectraLight QC) in a completely darkened room (0 lx as measured by the CL-500A), where it was exposed to a CIE standard illuminant D65, a CIE standard illuminant A, a white LED source and a fluorescent lamp (CWL) with two long-pass filters (Edmund Optics OD2 Longpass Filter, cut-on 600 nm and cut-on 500nm) between the device's optical aperture and the light source to ensure that all wavelengths of light from 200-600nm are rejected. Then the measurements were plotted, normalized to the area under the curve (Fig. 4).

The response below 500 nm, near the cut-on wavelength of the filter is due to stray light, which can be represented as a first approximation by a single stray-light coefficient [22]. The stray-light signal level, Y_0 is given by:

$$Y_0 = \varsigma_1 \sum_i Y(\lambda_i)\Delta\lambda, \tag{4}$$

where ς_1 is the stray-light coefficient, $Y(\lambda_i)$ is the signal at wavelength λ_i, and $\Delta\lambda$ is the wavelength interval. The stray-light coefficient is effectively the same for all test sources at 4.2×10^{-4} nm^{-1}. A limitation of this correction method is that it is constant for all wavelengths. However, as can be seen from Fig. 4, exposure of the device to light sources that contain a significant portion of power in the longer wavelengths (such as the halogen and the D65 light sources) elicits a

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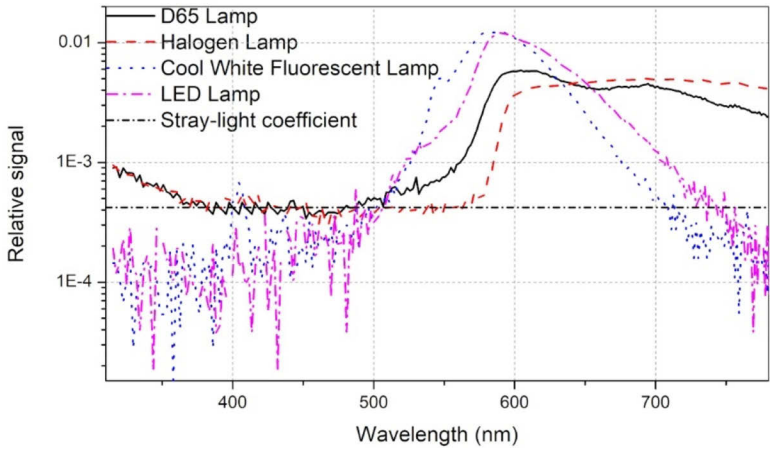


Fig. 4. Spectral response of the sensor (normalized to the area under each spectrum) measured with a long pass filter placed on the sensor.

stray-light response in the shorter wavelengths that is not uniform across wavelengths. This is due to stray-light from light outside the spectral range of the sensor. To confirm this phenomenon, namely to ensure that this effect observed is not due to the noise of the electronics, internal sensor optics or wavelength of light within the measurement range, the device was exposed to a halogen light source (Avantes AvaLight HAL-CAL) that includes NIR (750-1095 nm) spectral components, with a set of long-pass filters to suppress the shorter wavelengths (Edmund Optics OD2 Longpass Filter, cut-on 400 nm, 450 nm, 500 nm, 550 nm, 600 nm, 700 nm). Measurements were made with the first step of stray-light correction in (4) and the results are shown in Fig. 5(a).

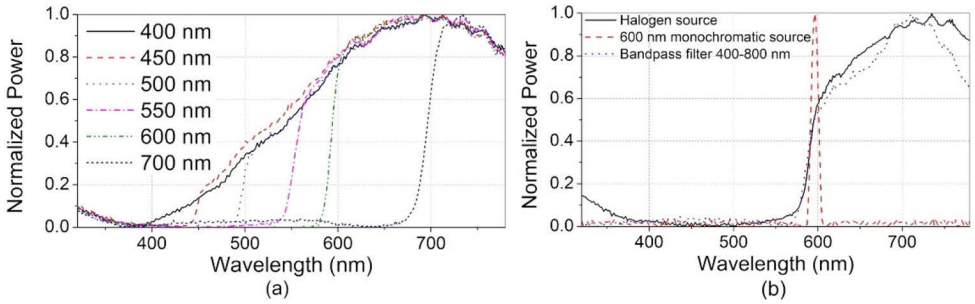


Fig. 5. (a) Spectral response of the device to halogen light measured with long pass filters with various cut-on wavelengths. (b) Spectral response of the device to a monochromatic (600 nm) light and halogen light measured with and without a 400-800 nm band pass filter.

The results indicate that there is a response at wavelengths below 400 nm due to stray light from light outside the spectral range of the sensor (> 780 nm). More importantly, the response is not uniform across the wavelengths below 400 nm, with an elevated level of stray light at shorter wavelengths. Another important observation is that the magnitude of the response due to stray-light is similar irrespective of the cut-on wavelength of the long-pass filter. This indicates that the stray-light response is due to wavelengths of light beyond the spectral range of the sensor. To confirm this, the above investigation was repeated with several modified measurement conditions. Firstly, another tungsten halogen light source (SpectraLight QC Illuminant A) with more powerful infrared content than the light used in Fig. 4 was used with a long-pass filter with

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cut-on wavelength of 600 nm to investigate if the relative response due to stray-light changes from what is illustrated in Fig. 5(a). Secondly, a monochromatic light source (600 nm) was used to investigate if it caused a stray-light response at shorter wavelengths. Finally, a bandpass filter of 400-800 nm was used with the halogen light source. Figure 5(b) shows the corresponding results, which confirm that the stray-light response is caused when the device is exposed to light sources with strong infrared content. Examples of such light sources are incandescent bulbs, halogen bulbs, and sunlight. The signal detected below the cut-on wavelength is due to stray-light and requires correction.

An obvious solution to eliminate the response due to stray-light is to use a low-pass filter to restrict the light that is incident on the device to wavelengths below 780 nm. However, using low-pass filters does not align with the requirement of the device to be robust and simple in design. Highly accurate methods of stray-light correction have been developed for effects towards the UV region [23–26] and hyperspectral imaging [27]. This device is intended for use in the 380 nm-780 nm wavelength range, and most of the response due to stray light lies outside this range. Furthermore, the stray-light response of the sensor would be negligible for most modern indoor light sources with almost no infrared content. We therefore opted to use a simple computational approach that uses the strength of longer wavelengths of visible light measured by the device as an indicator of the strength of the infrared content of the light source being measured. Using this approach, we predict the corresponding stray-light response of the sensor for each wavelength from 380 nm to 400 nm (the range within which the stray-light effect is significant).

We express the corresponding equations describing the stray-light calibration factor as:

$$\varsigma_2(\lambda_i) = m(\lambda_i)Y(780) + c(\lambda_i) \text{ for } 380\text{nm}<\lambda<400\text{nm}, \tag{5}$$

where $Y(\lambda_i)$ is the output of the device before stray-light correction, $m(\lambda_i) = a_1{\lambda_i}^4 + a_2{\lambda_i}^3 + a_3{\lambda_i}^2 + a_4\lambda_i + a_5$, $c(\lambda_i) = b_1{\lambda_i}^4 + b_2{\lambda_i}^3 + b_3{\lambda_i}^2 + b_4\lambda_i + b_5$ and a_{1-5}, b_{1-5} are device-dependent calibration parameters. The stray-light corrected output is given by:

$$Y_s(\lambda_i) = \frac{Y(\lambda_i) - Y_0}{\varsigma_2(\lambda_i)} \text{ for } 380 \text{ nm}<\lambda<400 \text{ nm}. \tag{6}$$

The result of the stray-light calibration performed on a halogen light source is shown in Fig. 6(a). The response that was observed at wavelengths below 400 nm, prior to the stray-light calibration is shown to be corrected.

2.2.6. Irradiance calibration

Characterization and calibration of the sensor allows for the determination the spectral responsivity $s(\lambda)$ of the device to enable the measurement of the SPD of an unknown input radiation incident on it, where $s(\lambda)$ is defined as:

$$s(\lambda) = \frac{Y(\lambda)}{X_\lambda(\lambda)}, \tag{7}$$

where $Y(\lambda)$ is the raw response from the sensor often measured as digital counts, and $X_\lambda(\lambda)$ is the SPD of the incident radiation.

Of the two methods used for spectrophotometer calibration, source-based calibration and detector-based calibration [22], we use the detector-based calibration method, which can be carried out using a standard spectrophotometer such as the CL-500A. With the sensor and the CL-500A placed under identical measurement conditions, the following definition from

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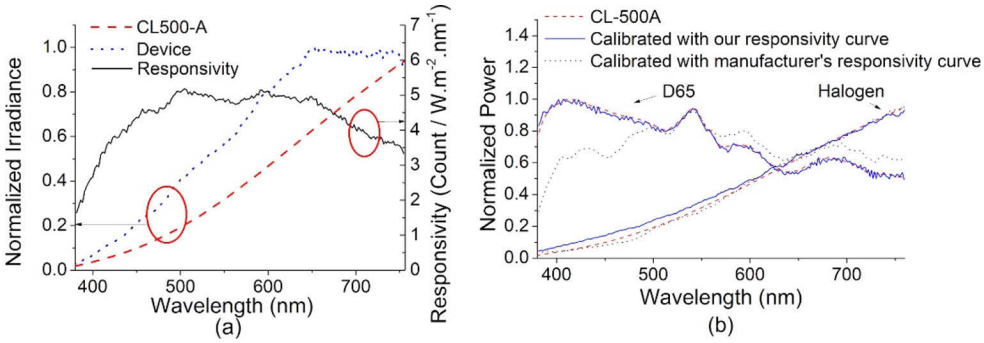


Fig. 6. (a) Responsivity of device calibrated against a standard spectrophotometer with a Halogen lamp. Measurements with the device were made after wavelength, baseline, stray-light, directional response and non-linearity calibrations are implemented. Normalized measurements of the halogen spectra taken by CL-500A and device are also shown. (b) Comparison of D65 and Halogen light calibrated with our responsivity curve, against the responsivity curve provided by the C12666MA's manufacturer.

CIE-233-2019 was simplified to determine the spectral responsivity of the test sensor:

$$s_{T,i} = s_{S,i}\frac{Y_{T,i}}{Y_{S,i}}.\frac{X_\lambda\,(\lambda_i,t_s)\prod_{j=1}^{n}a_{S,j}(\lambda_i,t_s)}{X_\lambda\,(\lambda_i,t_T)\prod_{j=1}^{n}a_{T,j}(\lambda_i,t_T)}, \tag{8}$$

where $S_{T,i}$ and $S_{S,i}$ are the responsivities, $Y_{T,i}$ and $Y_{S,i}$ are the signals, $X_\lambda\,(\lambda_i,t_T)$ and $X_\lambda\,(\lambda_i,t_S)$ are the SPDs of the radiation quantity at time t_S and t_T, $a_{S,j}(\lambda_i,t_S)$ and $a_{S,j}(\lambda_i,t_T)$ are the spectrum-dependent adjustment factors for the measurement at time t_S and t_T, for the i-th pixel of the test (T) and standard spectrophotometer (S) respectively.

If the measurement time and measurement position between both sensors is close enough, the radiation quantity $X_\lambda\,(\lambda_i)$ can be cancelled, which means that the radiometric absolute value does not have to be known if the test sensor's measurement application is similar to that of the standard spectrophotometer. The spectrum-dependent adjustment factors are also cancelled since both the sensor and the spectrophotometer have the same characteristics. The nonlinearity, baseline measurements, wavelength mapping, directional response, and stray-light behavior must be known and corrected such that both sensors' spectrum-dependent adjustment factors are sufficiently similar to simplify the spectral calibration process. This is ensured by the characterization processes described above for the test device. Therefore, assuming that $s_{S,i} = 1$ for all i for the standard spectrophotometer, the spectral responsivity of the test sensor can be simplified to be:

$$S_{T,i} = \frac{Y_{T,i}}{Y_{S,i}}. \tag{9}$$

To determine $s_{T,i}$, the sensor was mounted at 175 mm from a halogen light source (Avantes AvaLight HAL-CAL) on an optical bench and signal measurements taken. This was then repeated with the standard spectrophotometer and SPD of the radiation recorded. The process was carried out in a dark room with photopic illuminance at 0 lx. Directional response, baseline signal and non-linearities are corrected from the test sensor response, followed stray light calibration. The spectral responsivity shown in Fig. 6(a) was determined using the response of the device and the response from a standard spectrophotometer for a halogen light source. The responsivity deviates from that declared by the manufacturer due to a combination of our calibration procedures, especially the stray-light compensation. The measured SPD of a halogen and a D65 light source calibrated with our responsivity did better on average than that of the manufacturer's as shown in Fig. 6 (b).

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A flow chart demonstrating the entire calibration process in chronological order is shown in Fig. 7. It illustrates how light incident on the optically modified (for directional response improvement) device is first subjected to baseline calibration and non-linearity calibration. Next, wavelength calibration is performed on the signal, which then undergoes stray-light and irradiance calibration. The resulting output is the SPD of the incident light which is used for the calculation of the α-opic EDIs.

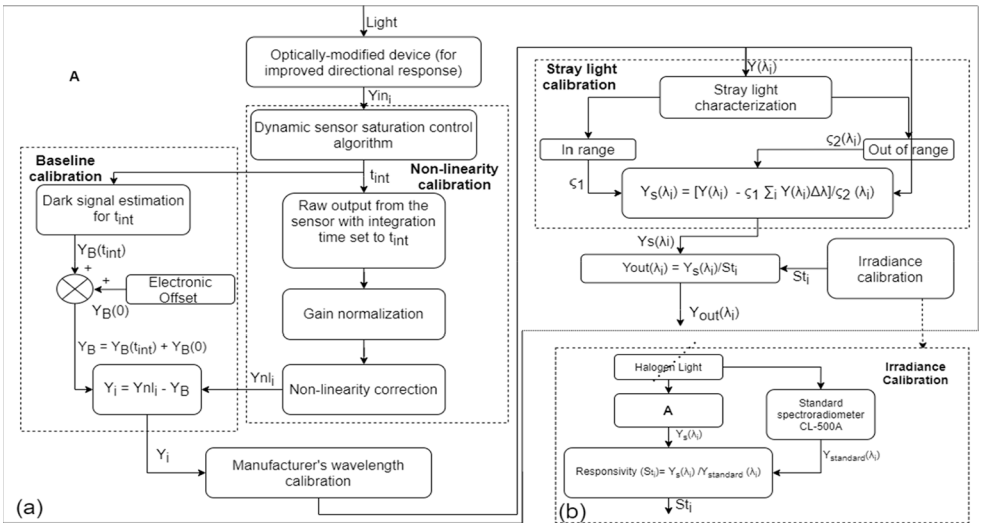


Fig. 7. Block diagram of the device calibration in chronological order (a) baseline, non-linearity, stray light calibration and (b) irradiance calibration.

3. Hardware, software, cloud sync and storage

The device was built around an nRF52832 System on Chip (SoC) from Nordic Semiconductors that permits floating-point operations on chip. This enables onboard processing of data to calculate the α-opic EDIs, while providing low power consumption and Bluetooth Low Energy (BLE) capabilities. Since the human circadian system is temporally sensitive, we included time-keeping functionality as well as onboard storage. The device includes 4Mb of internal storage which can hold data recorded for 5 days at the rate of 1 measurement per minute. A 110mAh 3.7 V Lithium polymer rechargeable battery provides this $44 \times 20 \times 29$ mm device weighing 20 gms, 24 hours of continuous operation on a single charge. A diagram of the device schematic and final product are shown in Fig. 1(b) and (c) respectively.

Using the measured SPD, the photopic and α-opic EDIs are calculated as per the metrics proposed by the CIE [1]. These values, in addition to the raw spectral data, are stored on board. We developed two interfaces for the device: a windows-based GUI and an Android app, which will allow the user to set device parameters (e.g. measurement interval), set the real time clock, and retrieve data. The Android app interface has the added functionality of uploading recorded data to the Cloud for further processing and analysis by the user.

4. Measurement accuracy, data collection and discussion

To demonstrate the accuracy and capabilities of the device as a practical measurement tool to be used in research and field applications, sensor data are presented here in several forms. Firstly, the accuracy of spectral calibration is presented followed by a demonstration of the data

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logging capability of the device. An investigation of the dynamic range of the device and the measurement repeatability of a single device and between multiple devices is also presented.

4.1. Spectral accuracy and data logging

Irradiances were measured under identical conditions for commonly encountered light sources (D65, halogen, fluorescent and LED) and the normalized root mean square error (NRMSE) was used to facilitate the comparison between SPDs. The associated quantitative measurement of errors in the α-opic EDIs, photopic illuminance and the NRMSE representing spectral accuracy are tabulated in Table 2. Since a warm-up time is hard to enforce in the intended application of the device, measurements were done at random times after switching on the device and the performance reported includes an average error from warm-up effects. Larger errors are observed for TL84 and LED sources in Fig. 8(b) at the longer and shorter wavelengths. The fact that a halogen lamp was used for irradiance calibration which also gives the most significant response for stray-light, can explain these errors in irradiance calibration of non-halogen light sources. However, even with the imperfections in the stray-light correction, the percentage difference between the α-opic EDI values calculated by using the irradiance spectra obtained by the device compared to a laboratory standard spectrophotometer was less than 7% for all light sources. This is less than the average percentage spectral mismatch errors of 15 light logging dosimeters, investigated by Price et. al. [15], for all of the α-opic and photopic sensitivity curves.

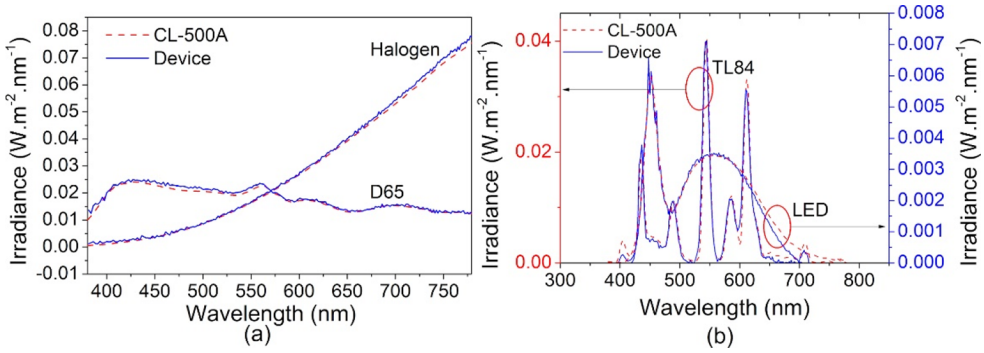


Fig. 8. Absolute irradiance measurement comparison of the device with the CL-500A standard spectrophotometer with the illuminant source as: (a) D65 and Halogen; (b) TL84 Fluorescent and LED.

Table 2. Percentage error of illuminances and spectral NRMSE for four different light sources measured by the device compared to CL-500A

Illuminance error (%)				
Illuminant	D65	A	TL84	LED
Photopic	0.23	2.24	4.70	0.17
Melanopic	1.80	2.51	0.77	0.67
L-Cone	1.87	2.20	4.04	0.27
S-Cone	1.06	4.36	6.96	3.25
M-Cone	0.13	2.39	4.68	0.81
Rhodopic	0.08	2.56	1.75	0.12
NRMSE	0.04	0.03	0.03	0.03

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The light sources tested have distinct spectral shapes, demonstrating the versatility of our device in accurately measuring the α-opic EDIs of all common forms of lighting. Samples of spectral irradiance data in terms of SPD are presented with their accuracy compared to a standard spectrophotometer in Fig. 8.

The data logging feature, where the device collects the spectral density data and computes illuminances for the light exposures of a test participant wearing the device at pre-set intervals, for a period of a day, is presented in Fig. 9. The participant was instructed to carry out normal daily routines and maintain normal sleep-wake times. The device was attached to the clothing of the chest of the participant (~20 cm from the chin) during waking hours and placed on a nearby surface for sleeping hours. Measurements were taken with a sampling frequency of 1 sample per minute.

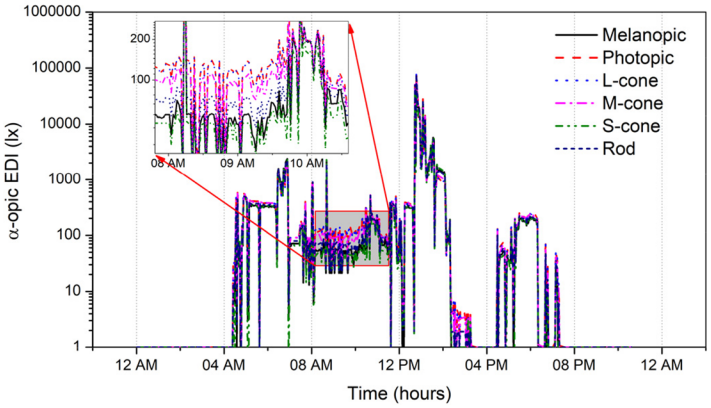


Fig. 9. The α-opic EDI data measured for an individual wearing the device for a day.

It is easily observed that the illuminance exposures of this participant in a typical day varies significantly in magnitude and relative to each other, further rationalizing the need to monitor them separately in studies sensitive to an individual's retinal photoreception.

4.2. Repeatability of measurements

Two forms of repeatability investigations were undertaken. Firstly, the repeatability within subsequent measurements of a single sensor device was investigated using a halogen light source. From the variations between 100 repeated measurements of SPD reported in Fig. 10, we observe that there is little variability within subsequent spectral measurements of a device. Although slightly amplified at higher irradiances (> 0.08 W.m^{-2}.nm^{-1}), it exhibits consistent performance in the mid-range 450-650 nm with irradiances in the range 0.01-0.08 W.m^{-2}.nm^{-1}. The largest variability was recorded at 780 nm at the highest irradiance at 0.12 ± 0.0009 W.m^{-2}.nm^{-1}. This is consistent with our findings reported in Fig. 3(b) which revealed greater deviation from linearity at higher irradiances. The intra-device repeatability (2σ) for photopic illuminance has a percentage error of 0.8% compared to 0.5% from the CL-500A.

Secondly, inter-device repeatability was investigated using 5 sensors for several common lighting conditions. The mean and standard deviation between devices for photopic illuminance and melanopic equivalent daylight illuminance (MEDI) were measured by placing the sensors in close proximity to each other. The measurements done for typically observed illuminance levels of sunlight and indoor LED lighting (employing tunable Phillips Hue bulbs) in a mock living room setup, are plotted against concurrent measurements taken by the standard spectrophotometer in Fig. 11.

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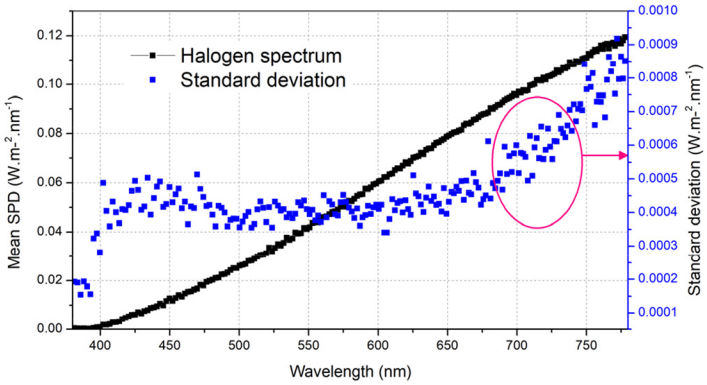


Fig. 10. Variation in repeated measurements (n=100) of a single device under halogen light.

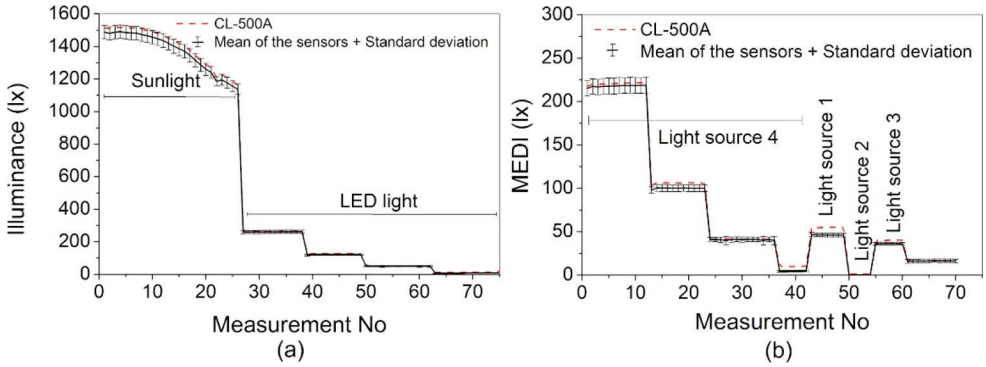


Fig. 11. Inter-device (n=5) variability in outdoor and LED indoor lighting (a) Photopic illuminance (b) MEDI for an arbitrary selection of LED spectra.

The results indicate that the maximum inter-device variability for photopic illuminance depicted in Fig. 11(a) is ± 40 lx at 1500 lx and ±8 lx at 250 lx, revealing low inter-device variations of 2.6% and 3.2% respectively for typical measurements. The MEDI measurements (Fig. 11(b)) taken under a set of four distinct spectral outputs (light sources 1-4) achieved using the tunable LEDs also show inter-device variability to be within 4% with ± 5 lx at 220 lx and ±2 lx at 50 lx. This is comparable to the inter-device repeatability reported by Raquel et. al. for the Kronowise ACM device [28].

4.3. Dynamic range

Measurements from three different light environments were taken to evaluate the dynamic range of the sensor device with concurrent measurements taken using CL-500A for comparison. First was a typical indoor artificial light (LED) environment where the photopic illuminance remained below 400 lx, followed by an indoor sun-lit day-time environment with illuminance up to 8000 lx and finally an outdoor environment with sunlight illuminance as high as 60,000 lx. From the measurements reported in Fig. 12, it can be seen that the sensor readings are in very good agreement with the CL-500A when exposed to four orders of illuminance levels across the tested light environments. Using a combination of calibration steps to correct for dark current, stray light, nonlinearity and the implementation of a dynamic sensor saturation control algorithm, we achieved a dynamic range of 1 lx to 60,000 lx with negligible saturation effects.

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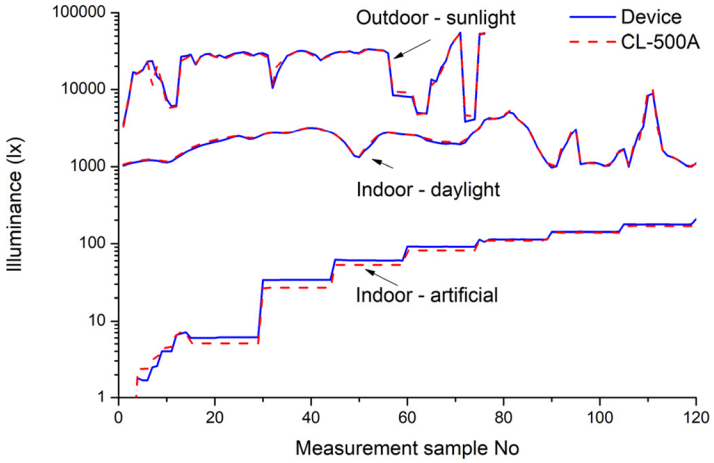


Fig. 12. Measurements in dynamic lighting environments demonstrating the dynamic range of the sensor, illustrated using of indoor environment with artificial LED lighting, indoor with daylight and outdoor environment with direct sunlight.

5. Conclusion

This device is among the first of its kind to report measurement accuracy for α-opic EDIs and photopic illuminance within 7.0% of the laboratory standard for four commonly encountered light sources and irradiance levels in our daily lives. This is a level of accuracy not seen before in a device with the size and usability of proposed wearable spectrophotometer. A directional response comparable to commercial illuminance meters was achieved with a dynamic range of 1-60,000 lx and maximum inter-device variability of 4%. The wavelength range of the device is between 380 and 780nm with an interpolated wavelength pitch of 5nm. Adaptations that were made to the calibration steps of lab scale spectrophotometers will enable the development of smaller, more targeted, cost-effective devices. Our device will serve as a high-fidelity research device from which precise predictions of the effect of light on human physiology can be made with an improved understanding of the metrology of optical radiation which targets human photoreception.

The device was designed, constructed, and calibrated for optimized measurements based on a mini-spectrophotometer with a resolution of 15 nm to last 24 hours on full charge and hold data measured for 5 days using 1-minute epochs. Data-driven approaches to simplify the calibration process of wearable spectral sensors will be explored in future work allowing us to take advantage of cheaper CMOS-based sensors.

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Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time due to privacy reasons, but may be obtained in future from the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

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